What is CSS?

- **CSS** stands for **Cascading Style Sheets.**
- CSS is used to define styles for your web pages, including the design, layout and variations in display for different devices and screen sizes.
- **CSS** describes how HTML elements are to be displayed on screen, paper, or in other media.

Benefits of CSS

- **CSS saves time** :
  - You can write CSS once and then reuse the same sheet in multiple HTML pages.
- **Easy maintenance**:
  - To make a global change, simply change the style, and all elements in all the web pages will be updated automatically.
- **Global web standards**:
  - It's a good idea to start using CSS in all the HTML pages to make them compatible with future browsers.
- **Platform Independence**:
  - The Script offer consistent platform independence and can support latest browsers as well.

CSS Comments

- Comments are used to explain the code, and may help when you edit the source code at a later date.
- Comments are ignored by browsers.
- A CSS comment is placed inside the <style> element, and starts with /* and ends with */
- Syntax:

```
/* This is CSS Comment Text */
```

Types Of CSS / Stylesheet

- CSS can be added to HTML documents in 3 ways:
  1. **Inline** - by using the style attribute inside HTML elements
  2. **Internal** - by using a <style> element in the <head> section
  3. **External** - by using a <link> element to link to an external CSS file
  4. **Multiple CSS**
- The most common way to add CSS, is to keep the styles in external CSS files.

1. Inline CSS :

- An inline CSS is used to apply a unique style to a single HTML element.
- An inline CSS uses the style attribute of an HTML element.
- Example:

```
<h1 style="color:blue;">A Blue Heading</h1>

<p style="color:red;">A red paragraph.</p>
```

2. Internal CSS :

- An internal CSS is used to define a style for a single HTML page.
- An internal CSS is defined in the <head> section of an HTML page, within a <style> element.
- Example :

```
<!DOCTYPE html>
<html>
<head>
    <style>
        body {background-color: powderblue;}
        h1   {color: blue;}
        p    {color: red;}
    </style>
</head>
```

```
<body>
    <h1>This is a heading</h1>
    <p>This is a paragraph.</p>
</body>
</html>
```

3. External CSS :

- An external style sheet is used to define the style for many HTML pages.
- To use an external style sheet, add a link to it in the <head> section of each HTML page:
- Example:

```
<!DOCTYPE html>
<html>
<head>
        <link rel="stylesheet" type="text/css" href="style.css">
</head>
<body>
        <h1>This is a heading</h1>
        <p>This is a paragraph.</p>
</body>
</html>
```

style.css

```
body
{
        background-color: powderblue;
}
h1
{
        color: blue;
}
p
{
        color: red;
}
```

4. Multiple CSS :

- Actually Multiple Style Sheet is not a type of Style Sheets but it is a combination of
    - External Style Sheet,
    - Internal Style Sheet and
    - Inline Style Sheet
        - as per designers requirement.
- We can use more then one style sheet in a HTML document file.
- This method of working with more then one style sheet that's why it is known as Multiple Style Sheet.
- Example:

```
<!DOCTYPE html>
<html>
<head>
        <link rel="stylesheet" type="text/css" href="style.css">
        <style>
                h1   {color: blue;}
                p    {color: red;}
        </style>
</head>
<body>
        <h1>This is a heading</h1>
        <p style="font-size:20px;" >This is a paragraph.</p>
</body>
</html>
```

style.css

```
body
{
        background-color: powderblue;
}
p
{
        background-color: yellow;
}
```

CSS Slectors

- There are two types of selector:
  1. Class
  2. ID

1. Class :

- The .class selector selects elements with a specific class attribute.
- To select elements with a specific class, write a period (.) character, followed by the name of the class.
- HTML elements can also refer to more than one class (look at Example 2 below).
- Syntax :

```
.class {
        css declarations;
}
```

- Example :

```
<html>
<head>
        <style>
                p.center {
                        text-align: center;
                        color: red;
                }
                p.large {
                        font-size: 30px;
                }
        </style>
</head>
<body>
        <p class="center large">This paragraph refers to two classes.</p>
</body>
</html>
```

2. ID :

- The #id selector styles the element with the specified id.
- Syntax :

```
#id {
        css declarations;
}
```

- Example :

```
<html>
<head>
        <style>
                #firstname {
                        font-size: 25px;
                        color: blue;
                }
        </style>
</head>
<body>
        <p id="firstname">This paragraph refers to two classes.</p>
</body>
</html>
```

CSS Font Properties

- The CSS font properties allow you to change the font family, boldness, size and the style of a text.

  [1] Font-family Property

  [2] Font-style Property

  [3] Font-weight Property

  [4] Font-size Property

  [5] Font-variant Property

1. Font-family :

- The font-family property specifies the font for an element.
- There are two types of font family names:
  - **family-name** - The name of a font-family, like "times", "courier", "arial", etc.
  - **generic-family** - The name of a generic-family, like "serif", "sans-serif", "cursive", "fantasy", "monospace".
- Syntax :

  font-family: *family-name|generic-family*

- Example :

```
<html>
<head>
        <style>
                p {
                        font-family: "calibri", Times, serif;
                }
        </style>
</head>
<body>
        <h1>The font-family Property</h1>
        <p>This is a paragraph, shown in the Times New Roman font.</p>
</body>
</html>
```

2. Font-style :

- The font-style property specifies the font style for a text.
- Syntax :

```
font-style: normal | italic | oblique;
```

- Example :

```
<html>
<head>
    <style>
        p {
            font-style: italic;
        }
    </style>
</head>
<body>
    <h1>The font-family Property</h1>
    <p>This is a paragraph, shown in the Times New Roman font.</p>
</body>
</html>
```

3. Font-weight :

- The font-weight property sets how thick or thin characters in text should be displayed.
- Syntax :

```
font-weight: normal | bold | bolder | lighter | number
```

- Example :

```
<html>
<head>
    <style>
        p {
            font-weight: bold;
        }
    </style>
</head>
<body>
    <p>This is a paragraph.</p>
</body>
</html>
```

4. Font-size :

- The font-size property sets the size of a font.
- Syntax :

  font-size : medium | xx-small | x-small | small | large | x-large | xx-large | smaller | larger | length | % | px

- Example :

```
<html>
<head>
        <style>
                p { font-size: xx-large; }
                font { font-size: xx-small; }
        </style>
</head>
<body>
        <p>This is a xx-large paragraph.</p>
        <font>This is a xx-small paragraph.</font>
</body>
</html>
```

5. Font-variant :

- In a small-caps font, all lowercase letters are converted to uppercase letters.
- Syntax :

  font-variant : normal | small-caps

- Example :

```
<html>
<head>
        <style>
                h1 { font-variant: small-caps; }
        </style>
</head>
<body>
        <h1> Play with the two different font variants!</h1>
</body>
</html>
```

CSS Text Properties

- The CSS text properties allow you to change the appearance of text.
- It is possible to change the color of text, increase or decrease the space between characters in a text, align a text, decorate a text, indent the first line in a text and more.

[1] Color Property

[2] Text-align Property

[3] Text-decoration Property

[4] Text-transform Property

[5] Word-spacing Property

[6] Letter-spacing Property

1. Color :

- The color property specifies the color of text.
- Syntax :

```
Color : colorname;
```

- Example :

```
<html>
<head>
    <style>
        body {  color: red;  }
        h1 { color: #00ff00;  }
        font { color: rgb(0,0,255);  }
    </style>
</head>
<body>
    <h1>This is heading 1</h1>

    <p>This is an ordinary paragraph. Notice that this text is red. The
    default text-color for a page is defined in the body selector.</p>

    <font>This is a paragraph with font. This text is blue.</font>
</body>
</html>
```

2. Text-align :

- The text-align property specifies the horizontal alignment of text in an element.
- Syntax :

  text-align: left | right | center | justify ;

- Example :

```
<html>
<head>
        <style>
                p { text-align: center;  }
        </style>
</head>
<body>

        <p>This is an ordinary paragraph. Notice that this text is red. The
        default text-color for a page is defined in the body selector.</p>

</body>
</html>
```

3. Text-decoration :

- The text-decoration property specifies the decoration added to text, and is a shorthand property for.
  - text-decoration-line (required)
  - text-decoration-color
  - text-decoration-style
- Syntax :

  text-decoration: [text-decoration-line] [text-decoration-color] [text-decoration-style];

- Example :

```
<html>
<head>
        <style>
                h1 {  text-decoration: overline; }
                h2 {  text-decoration: line-through; }
                h3 {  text-decoration: underline; }
                h4 {  text-decoration: underline overline; }
```

```
        </style>
</head>
<body>

        <h1>This is heading 1</h1>
        <h2>This is heading 2</h2>
        <h3>This is heading 3</h3>
        <h4>This is heading 4</h4>

</body>
</html>
```

4. Text-transform :

- The text-transform property controls the capitalization of text.
- Syntax :

  text- transform : none | capitalize | uppercase | lowercase

- Example :

```
<html>
<head>
        <style>
                h1 {  text-transform : capitalize; }
                h2 {  text-transform : uppercase; }
                h3 {  text-transform : lowercase; }
        </style>
</head>
<body>

        <h1>This is capitalize text.</h1>
        <h2>This is uppercase text.</h2>
        <h3>This is lowercase.</h3>

</body>
</html>
```

5. Word-spacing :

- The word-spacing property increases or decreases the white space between words.
- Syntax :

  text- spacing : normal | *length*

- Example :

```
<html>
<head>
        <style>
                p {  word-spacing: 30px; }
        </style>
</head>
<body>
        <p>This is some text. This is some text.</p>
</body>
</html>
```

6. letter-spacing :

- The letter-spacing property increases or decreases the space between characters in a text.
- Syntax :

  Letter-spacing : normal | *length*

- Example :

```
<html>
<head>
    <style>
            p {  letter-spacing: 5px; }
    </style>
</head>
<body>
    <p>This is some text. This is some text.</p>
</body>
</html>
```

CSS Background Properties

- The CSS Background properties allow you to control the background color of an element.
- Set an image as the background, repeat a background image, vertically or horizontally and position an image on a page.

  [1] Background-color Property

  [2] Background-image Property

  [3] Background-repeat Property

  [4] Background-attachment Property

  [5] Background-position Property

1. Background-color :

- The background-color property sets the background color of an element.
- Syntax :

  background-color: color | transparent;

- Example :

```
<html>
<head>
        <style>
                body {  background-color: orange;  }   OR
                body {  background-color: #ff9900;  }  OR
                body {  background-color: rgb(255,130,255);  }
        </style>
</head>
<body>

        <h1>The background-color Property</h1>
        <p>The background color can be specified with a color name.</p>

</body>
</html>
```

2. Background-image :

- The background-image property sets one or more background images for an element.
- By default, a background-image is placed at the top-left corner of an element, and repeated both vertically and horizontally.
- Syntax :

    background-image: URL | none;

- Example :

```
<html>
<head>
        <style>
                body {  background-color: url("nature.jpg");  }
        </style>
</head>
<body>

        <h1>The background-image Property</h1>
        <p>Hello World!</p>

</body>
</html>
```

3. Background-repeat :

- The background-repeat property sets if/how a background image will be repeated.
- By default, a background-image is repeated both vertically and horizontally.
- Syntax :

    background-repeat: repeat | repeat-x | repeat-y | no-repeat;

Value	Description
repeat	The background image is repeated both vertically and horizontally
repeat-x	The background image is repeated only horizontally
repeat-y	The background image is repeated only vertically
no-repeat	The background-image is not repeated. The image will only be shown once

- Example :

```
<html>
<head>
        <style>
                body {
                        background-color: url("pattern.jpg");
                        background-repeat: no-repeat;
                        }
        </style>
</head>
<body>

        <h1>The background-repeat Property</h1>
        <p> Here, the background image is repeated only vertically.</p>

</body>
</html>
```

4. **Background-attachment :**

- The background-attachment property sets whether a background image scrolls with the rest of the page, or is fixed.
- Syntax :

```
background-attachment: scroll | fixed;
```

Value	Description
scroll	The background image will scroll with the page. This is default
fixed	The background image will not scroll with the page

- Example :

```
<html>
<head>
        <style>
                body {
                        background-color: url("pattern.jpg");
                        background-repeat: no-repeat;
                        background-attachment: fixed;
                        }
```

```
        </style>
</head>
<body>

        <h1>The background-attachment Property</h1>
        <p> The background-image is fixed. Try to scroll down the
page..</p>
        .
        .
        .
        <p> The background-image is fixed. Try to scroll down the
page..</p>

</body>
</html>
```

5. **Background-position :**

- The background-position property sets the starting position of a background image.
- Syntax :

```
background-position: value;
```

Value	Description
left top **left center** **left bottom** **right top** **right center** **right bottom** **center top** **center center** **center bottom**	If you only specify one keyword, the other value will be "center"
x% y%	The first value is the horizontal position and the second value is the vertical. The top left corner is 0% 0%.
xpos ypos	The first value is the horizontal position and the second value is the vertical. The top left corner is 0 0.

- Example :

```
<html>
<head>
    <style>
        body {
            background-color: url("pattern.jpg");
            background-repeat: repeat;
            background-position: center bottom;
            }
    </style>
</head>
<body>

    <h1>The background-repeat Property</h1>
    <p> Here, the background image is repeated only vertically.</p>

</body>
</html>
```

CSS List-style Properties

- The list-style property is a shorthand for the following properties:
    - list-style-type
    - list-style-position
    - list-style-image
- Syntax :

    list-style: *list-style-type list-style-position list-style-image;*

1. **List-style-type :**
    - The list-style-type specifies the type of list-item marker in a list.
    - Syntax :

        list-style-type: value;

Value	Description
none	No marker is shown
disc	Default value. The marker is a filled circle
circle	The marker is a circle
square	The marker is a square
decimal	The marker is a number
lower-alpha	The marker is lower-alpha (a, b, c, d, e, etc.)
lower-greek	The marker is lower-greek
lower-latin	The marker is lower-latin (a, b, c, d, e, etc.)
lower-roman	The marker is lower-roman (i, ii, iii, iv, v, etc.)
upper-alpha	The marker is upper-alpha (A, B, C, D, E, etc.)
upper-greek	The marker is upper-greek
upper-latin	The marker is upper-latin (A, B, C, D, E, etc.)
upper-roman	The marker is upper-roman (I, II, III, IV, V, etc.)
almenian	The marker is traditional Armenian numbering

- Example :

```
<html>
<head>
        <style>
                ul {
                        list-style-type: decimal;
                }
        </style>
</head>
<body>
        <ul>
                <li>Coffee</li>
                <li>Tea</li>
                <li>Coca Cola</li>
        </ul>
</body>
</html>
```

2. List-style-position :

- The list-style-position property specifies the position of the list-item markers (bullet points).
- Syntax :

```
list-style-position: inside | outside;
```

1. inside : The bullet points will be inside the list item
2. outside : The bullet points will be outside the list item. This is default

- Example :

```
<html>
<head>
        <style>
                ul {
                        list-style-position: inside;
                }
        </style>
</head>
<body>
```

```
        <ul>
                <li>Coffee</li>
                <li>Tea</li>
                <li>Coca Cola</li>
        </ul>
</body>
</html>
```

3. List-style-image :

- The list-style-image property replaces the list-item marker with an image.
- Syntax :

```
list-style-image: none | URL;
```

- Example :

```
<html>
<head>
        <style>
                ul {
                        list-style-image: url('square.png');
                }
        </style>
</head>
<body>
        <ul>
                <li>Coffee</li>
                <li>Tea</li>
                <li>Coca Cola</li>
        </ul>
</body>
</html>
```

CSS Padding Properties

- An element's padding is the space between its content and its border.
- This property can have from one to four values.
    - **If the padding property has four values:**
        - padding:10px 5px 15px 20px;
            - top padding is 10px
            - right padding is 5px
            - bottom padding is 15px
            - left padding is 20px
    - **If the padding property has three values:**
        - padding:10px 5px 15px;
            - top padding is 10px
            - right and left padding are 5px
            - bottom padding is 15px
    - **If the padding property has two values:**
        - padding:10px 5px;
            - top and bottom padding are 10px
            - right and left padding are 5px
    - **If the padding property has one value:**
        - padding:10px;
            - all four paddings are 10px
    - padding-top: 10px;
    - padding-bottom: 10px;
    - padding-left: 10px;
    - padding-right: 10px;
- **Note:** Negative values are not allowed.

- Syntax:

```
padding: length;
```

- *length* - specifies a padding in px, pt, cm, etc.
- *%* - specifies a padding in % of the width of the containing element.

- Example:

```
<html>
<head>
        <style>
                p {
                        padding: 70px;
                }
        </style>
</head>
<body>
        <p>
                This element has a padding of 70px.
        </p>
</body>
</html>
```

CSS Margin Properties

- The CSS margin properties are used to create space around elements, outside of any defined borders.
- This property can have from one to four values.
  - **If the margin property has four values:**
    - margin :10px 5px 15px 20px;
      - top margin is 10px
      - right margin is 5px
      - bottom margin is 15px
      - left margin is 20px
  - **If the margin property has three values:**
    - margin :10px 5px 15px;
      - top margin is 10px
      - right and left margin are 5px
      - bottom margin is 15px
  - **If the margin property has two values:**
    - margin :10px 5px;
      - top and bottom margin are 10px
      - right and left margin are 5px
  - **If the margin property has one value:**
    - margin :10px;
      - all four margin are 10px
  - margin-top: 10px;
  - margin-bottom: 10px;
  - margin-left: 10px;
  - margin-right: 10px;
- **Note:** Negative values are allowed.

- Syntax:

  margin: length | auto;

  - *length* - specifies a padding in px, pt, cm, etc.
  - *%* - specifies a padding in % of the width of the containing element.
  - *Auto* - The browser calculates a margin

- Example:

```
<html>
<head>
    <style>
        p {
            margin: 30px;
        }
    </style>
</head>
<body>
    <p>
        This element has a margin of 30px.
    </p>
    <p>
        This element has a margin of 30px.
    </p>
</body>
</html>
```

CSS Border Properties

- The CSS border properties allow you to specify the style, width, and color of an element's border.
- The border property is a shorthand property for:
    - border-width
    - border-style (required)
    - border-color
- Syntax :

  border: border-width border-style border-color;

1. **Border-width :**

- The border-width property sets the width of an element's four borders. This property can have from one to four values.
- The width can be set as a specific size (in px, pt, cm, em, etc) or by using one of the three pre-defined values: thin, medium, or thick.
- Syntax:

  border-width: medium | thin | thick | *length*;

- **If the border-style property has four values:**
    - border-width: thin medium thick 10px;
        - top border is thin
        - right border is medium
        - bottom border is thick
        - left border is 10px
- **If the border-style property has three values:**
    - border-width: thin medium thick;
        - top border is thin
        - right and left borders are medium
        - bottom border is thick

- **If the border-style property has two values:**
  - border-width: thin medium;
    - top and bottom borders are thin
    - right and left borders are medium
- **If the border-style property has one values:**
  - border-width: thin;
    - all four borders are thin
- border-top-width: 10px;
- border-bottom-width: 10px;
- border-left-width: 10px;
- border-right-width: 10px;
- Example : **HTML Code :**

```
<html>
<head>
        <style>
                p {
                        border-style: solid;
                        border-width: thin 20px medium;
                }
        </style>
</head>
<body>

        <p>Some Text Here.</p>

</body>
</html>
```

2. Border-style :

- The border-style property sets the style of an element's four borders. This property can have from one to four values.
- Syntax:

  border-style: none | hidden | dotted | dashed | solid | double | groove | ridge | inset | outset;

dotted	Defines a dotted border
dashed	Defines a dashed border
solid	Defines a solid border
double	Defines a double border
groove	Defines a 3D grooved border. The effect depends on the border color value
ridge	Defines a 3D ridged border. The effect depends on the border color value
inset	Defines a 3D inset border. The effect depends on the border color value
outset	Defines a 3D outset border. The effect depends on the border color value
none	Defines no border
hidden	Defines a hidden border

- **If the border-style property has four values:**
  - border-style: dotted solid double dashed;
    - top border is dotted
    - right border is solid
    - bottom border is double
    - left border is dashed
- **If the border-style property has three values:**
  - border-style: dotted solid double;
    - top border is dotted
    - right and left borders are solid
    - bottom border is double
- **If the border-style property has two values:**
  - border-style: dotted solid;
    - top and bottom borders are dotted
    - right and left borders are solid

- **If the border-style property has one values:**
  - border-style: dotted;
    - all four borders are dotted
- border-top-style: dotted;
- border-bottom- style: dashed;
- border-left- style: double;
- border-right- style: solid;

- Example : **HTML Code :**

```
<html>
<head>
        <style>
                p {
                        border-style: dotted;
                }
        </style>
</head>
<body>

        <p>Some Text Here.</p>

</body>
</html>
```

3. Border-color :

- The border-color property is used to set the color of the four borders.
- The color can be set by:
  - ✓ name - specify a color name, like "red"
  - ✓ HEX - specify a HEX value, like "#ff0000"
  - ✓ RGB - specify a RGB value, like "rgb(255,0,0)"
  - ✓ transparent
- Syntax:

  border-color: color | transparent;

- **If the border-color property has four values:**
  - border-color: red green blue pink;

        - top border is red
        - right border is green
        - bottom border is blue
        - left border is pink
- **If the border-color property has three values:**
    - border-color: red green blue;
        - top border is red
        - right and left borders are green
        - bottom border is blue
- **If the border-color property has two values:**
    - border-color: red green;
        - top and bottom borders are red
        - right and left borders are green
- **If the border-color property has one value:**
    - border-color: red;
        - all four borders are red
- border-top-color: red;
- border-bottom- color: green;
- border-left- color: blue;
- border-right- color: cyan;
- Example : **HTML Code :**

```
<html>
<head>
        <style>
                p {
                        border-color: red;
                }
        </style>
</head>
<body>
        <p>Some Text Here.</p>
</body>
</html>
```

CSS3 Backgound-Gradients Properties

- CSS3 gradients let you display smooth transitions between two or more specified colors.
- CSS3 defines two types of gradients:
    - **Linear Gradients** (goes down/up/left/right/diagonally)
    - **Radial Gradients** (defined by their center)

1. **Linear Gradients:**

- To create a linear gradient you must define at least two color stops.
- Color stops are the colors you want to render smooth transitions among.
- You can also set a starting point and a direction (or an angle) along with the gradient effect.
- Syntax :

```
background: linear-gradient(direction, color-stop1, color-stop2, ...);
```

- Example :

```
<html>
<head>
        <style>
                div { height: 300px; }
                div.simple {  background: linear-gradient(red, yellow);  }
                div.side {  background: linear-gradient(to right, red , yellow); }
                div.angle {  background: linear-gradient(45deg, red , yellow); }
        </style>
</head>
<body>
        <div class="simple">Some Text Here.</div>
        <div class="side">Some Text Here.</div>
        <div class="angle">Some Text Here.</div>
</body>
</html>
```

2. **Radial Gradients:**

- A radial gradient is defined by its center.
- To create a radial gradient you must also define at least two color stops.
- Syntax :

  background: radial-gradient(*shape size* at *position, start-color, ..., last-color*);

- Example :

```
<html>
<head>
        <style>
                div { height: 300px; }
                div.simple {  background: radial-gradient(red, yellow, green);  }
                div.stops {  background: radial-gradient(red 5%, yellow 15%, green
                60%) ; }
                div.shape {  background: radial-gradient(circle, red , yellow, green);
                }
        </style>
</head>
<body>
        <div class=”simple”>Some Text Here.</div>
        <div class=”stops”>Some Text Here.</div>
        <div class=”shape”>Some Text Here.</div>
</body>
</html>
```

CSS3 Shadow Properties

- With CSS3 you can add shadow to text and to elements.
- In this chapter you will learn about the following properties:
    1. **text-shadow**
    2. **box-shadow**

1. **text-shadow:**

- The CSS3 text-shadow property applies shadow to text.
- Syntax :

```
text-shadow: x  y  blur  color;
```

- Example :

```
<html>
<head>
        <style>
                div.simple {  text-shadow: 0 0 3px #FF0000;  }
                div.multiple {  text-shadow: 0 0 3px #FF0000, 0 0 5px #0000FF; }
        </style>
</head>
<body>
        <div class=”simple”>Some Text Here.</div>
        <div class=”multiple”>Some Text Here.</div>
</body>
</html>
```

2. **box-shadow:**

- The CSS3 box-shadow property applies shadow to elements.
- Syntax :

```
box-shadow: x  y  blur  color;
```

- Example :

```
<html>
<head>
        <style>
                div {
                        height: 300px; box-shadow: 10px 10px 5px gray;  }
        </style>
</head>
```

```
<body>
	<div>Some Text Here.</div>
</body>
</html>
```

CSS3 2D/3D Transforms Properties

- CSS3 transforms allow you to translate, rotate, scale, and skew elements.
- A transformation is an effect that lets an element change shape, size and position.
- CSS3 supports 2D and 3D transformations.
    1. **translate()**
    2. **rotate() => rotateX(),rotateY()**
    3. **scale()**
    4. **skew()**

1. **translate:**

- The translate() method moves an element from its current position (according to the parameters given for the X-axis and the Y-axis).
- Syntax :

```
transform: translate(x, y);
```

- Example :

```
<html>
<head>
	<style>
		div {
			width: 200px;
			height: 200px;
			background-color: red;
			transform: translate(50px, 100px);  }
	</style>
</head>
<body>
	<div>Transformed Element</div>
</body>
</html>
```

2. **rotate:**

- The rotate() method rotates an element clockwise or counter-clockwise according to a given degree.
- Syntax :

```
transform: rotate(deg);
```

- Example :

```
<html>
<head>
        <style>
                div {
                        width: 200px;
                        height: 200px;
                        background-color: red;
                        transform: rotate(45deg);  }
        </style>
</head>
<body>
        <div>Transformed Element</div>
</body>
</html>
```

3. **Scale:**

- The scale() method increases or decreases the size of an element (according to the parameters given for the width and height).
- Syntax :

```
transform: scale(x, y);
```

- Example :

```
<html>
<head>
        <style>
                div {
                        width: 200px;
                        height: 200px;
                        background-color: red;
                        transform: scale(2, 3);  }
```

```
        </style>
    </head>
    <body>
        <div>Transformed Element</div>
    </body>
    </html>
```

4. **skew:**

- The skew() method skews an element along the X-axis and Y-axis by the given angle.
- Syntax :

```
transform: skew(x, y);
```

- Example :

```
<html>
<head>
    <style>
        div {
            width: 200px;
            height: 200px;
            background-color: red;
            transform: skew(20deg, 10deg); }
    </style>
</head>
<body>
    <div>Transformed Element</div>
</body>
</html>
```

CSS3 Transition Properties

- To create a transition effect, you must specify two things:
  - the CSS property you want to add an effect to
  - the duration of the effect
- Example :

```
<!DOCTYPE html>
<html>
<head>
<style>
        .div1 {
           width: 100px;
           height: 100px;
           background: red;
           transition: width 2s;
        }
        .div1:hover {
           width: 300px;
        }
</style>
</head>
<body>

        <div class="div1"> Hover over the div element above, to see the
        transition effect.</div>

</body>
</html>
```

CSS3 Position Properties

- The position property specifies the type of positioning method used for an element.
- There are four different position values:
  - **static**
  - **relative**
  - **fixed**
  - **absolute**
- **Example:**
  - HTML elements are positioned static by default.
  - Static positioned elements are not affected by the top, bottom, left, and right properties.

```
<!DOCTYPE html>
<html>
<head>
<style>
        div.static {
                border: 2px green solid;
                position: static;
        }
        div.relative {
                border: 2px green solid;
                position: relative;
                left: 50px;
        }
        div.fixed {
                border: 2px green solid;
                position: fixed;
                right: 50px;
                bottom: 50px;
        }
        div.absolute {
                border: 2px green solid;
                position: absolute;
                top: 100px;
                left: 200px;
```

```
        }

</style>
</head>
<body>

        <div class="static"> This div element has position: static;</div>
        <div class="relative"> This div element has position: relative;</div>
        <div class="fixed"> This div element has position: static;</div>
        <div class="absolute"> This div element has position: static;</div>

</body>
</html>
```